nature methods

BRIEF COMMUNICATION

https://doi.org/10.1038/s41592-019-0615-4

Light-sheet microscopy of cleared tissues with isotropic, subcellular resolution

Tonmoy Chakraborty[1,2], Meghan K. Driscoll[2], Elise Jeffery[3], Malea M. Murphy[3], Philippe Roudot[2], Bo-Jui Chang[1,2], Saumya Vora[1], Wen Mai Wong[4,5], Cara D. Nielson[4,5], Hua Zhang[6], Vladimir Zhemkov[6], Chitkale Hiremath[1,7], Estanislao Daniel De La Cruz[8], Yating Yi[9], Ilya Bezprozvanny[6], Hu Zhao[9], Raju Tomer[8,10,11], Rainer Heintzmann[12,13], Julian P. Meeks[4,5], Denise K. Marciano[1,7], Sean J. Morrison[3,14,15], Gaudenz Danuser[1,2], Kevin M. Dean[1*] and Reto Fiolka[1,2*]

We present cleared-tissue axially swept light-sheet microscopy (ctASLM), which enables isotropic, subcellular resolution imaging with high optical sectioning capability and a large field of view over a broad range of immersion media. ctASLM can image live, expanded, and both aqueous and non-aqueous chemically cleared tissue preparations. Depending on the optical configuration, ctASLM provides up to 260 nm of axial resolution, a three to tenfold improvement over confocal and other reported cleared-tissue light-sheet microscopes. We imaged millimeter-scale cleared tissues with subcellular three-dimensional resolution, which enabled automated detection of multicellular tissue architectures, individual cells, synaptic spines and rare cell–cell interactions.

Human tissues are composed of multiple polarized cell types organized in well-defined three-dimensional (3D) architectures. Interestingly, it has been shown that rare subsets of cells play a crucial role in disease progression[1] and interdisciplinary efforts now aim to generate comprehensive atlases of human cells in diverse tissue types. To date, this has largely relied on massively parallel sequencing and machine-learning-based analyses to identify unique subpopulations of cells. Combined with advanced imaging, such efforts could not only shed light on the diversity of cell types, but the biological context in which each population operates. However, imaging large tissues with subcellular resolution remains challenging owing to the heterogeneous refractive index and composition of tissues, which results in complex aberrations and an increased scattering coefficient, both of which decrease spatial resolution and limit the depth of imaging[2].

To overcome this challenge, a large variety of clearing techniques have been developed that aim to homogenize the optical properties of the tissue using aqueous or organic solvents[3]. These clearing techniques routinely render biological specimens sufficiently transparent, such that, if the imaging technology was to exist, entire organisms could be imaged with subcellular resolution and with molecular specificity. Because the specimens are 3D, an ideal imaging system should possess isotropic resolution to accurately measure 3D cellular morphology, biomolecular localization and signaling activity[4]. Nevertheless, imaging chemically cleared specimens with diffraction-limited or super resolution presents technical challenges. For example, each clearing mechanism provides advantages and disadvantages and requires unique immersion media with refractive indices that range between 1.33 and 1.559. Thus, the imaging system must optimally operate throughout this refractive index range without suffering from deleterious aberrations that decrease resolution and sensitivity.

Light-sheet fluorescence microscopy (LSFM), because of its parallelized image acquisition, inherent optical sectioning and ability to image large biological structures quickly and with high optical resolution, serves as an ideal candidate for cleared tissue imaging (Supplementary Table 1)[5]. Nevertheless, to the best of our knowledge, there is not yet a light-sheet microscope for cleared tissue specimens that possesses submicron, isotropic resolution and is also compatible with the full range of clearing methods[6–9]. Although lattice light-sheet microscopy readily achieves high spatial resolution, it relies on specialized high numerical aperture lenses that are only compatible with aqueous solvents (refractive index = 1.333) and its field of view is relatively small. Here, we address these limitations and present a scalable imaging platform that provides subcellular anatomical detail in any spatial dimension across millimeter-wide cubes of tissue. The system that we refer to as ctASLM, builds on previous work where the sample was mechanically scanned through a thin light-sheet[10,11]. Instead of scanning samples slowly, ctASLM scans the illumination in its propagation direction using fast aberration-free remote focusing (Supplementary Note 1 and Supplementary Fig. 1), refractive index-independent illumination and detection optics, residual-spherical-aberration correction in

[1]Department of Cell Biology, University of Texas Southwestern Medical Center, Dallas, TX, USA. [2]Lyda Hill Department of Bioinformatics, University of Texas Southwestern Medical Center, Dallas, TX, USA. [3]Children's Research Institute, University of Texas Southwestern Medical Center, Dallas, TX, USA. [4]Department of Neuroscience, University of Texas Southwestern Medical Center, Dallas, TX, USA. [5]Department of Neurology and Neurotherapeutics, University of Texas Southwestern Medical Center, Dallas, TX, USA. [6]Department of Physiology, University of Texas Southwestern Medical Center, Dallas, TX, USA. [7]Department of Internal Medicine, University of Texas Southwestern Medical Center, Dallas, TX, USA. [8]Department of Biological Sciences, Columbia University, New York, NY, USA. [9]Department of Restorative Sciences, Texas A&M University, College Station, TX, USA. [10]NeuroTechnology Center, Columbia University, New York, NY, USA. [11]Data Science Institute, Columbia University, New York, NY, USA. [12]Leibniz Institute of Photonic Technology, Jena, Germany. [13]Institute of Physical Chemistry and Abbe Center of Photonics, Friedrich Schiller University, Jena, Germany. [14]Department of Pediatrics, University of Texas Southwestern Medical Center, Dallas, TX, USA. [15]Howard Hughes Medical Institute, University of Texas Southwestern Medical Center, Dallas, Texas, USA. *e-mail: Kevin.Dean@utsouthwestern.edu; Reto.Fiolka@utsouthwestern.edu

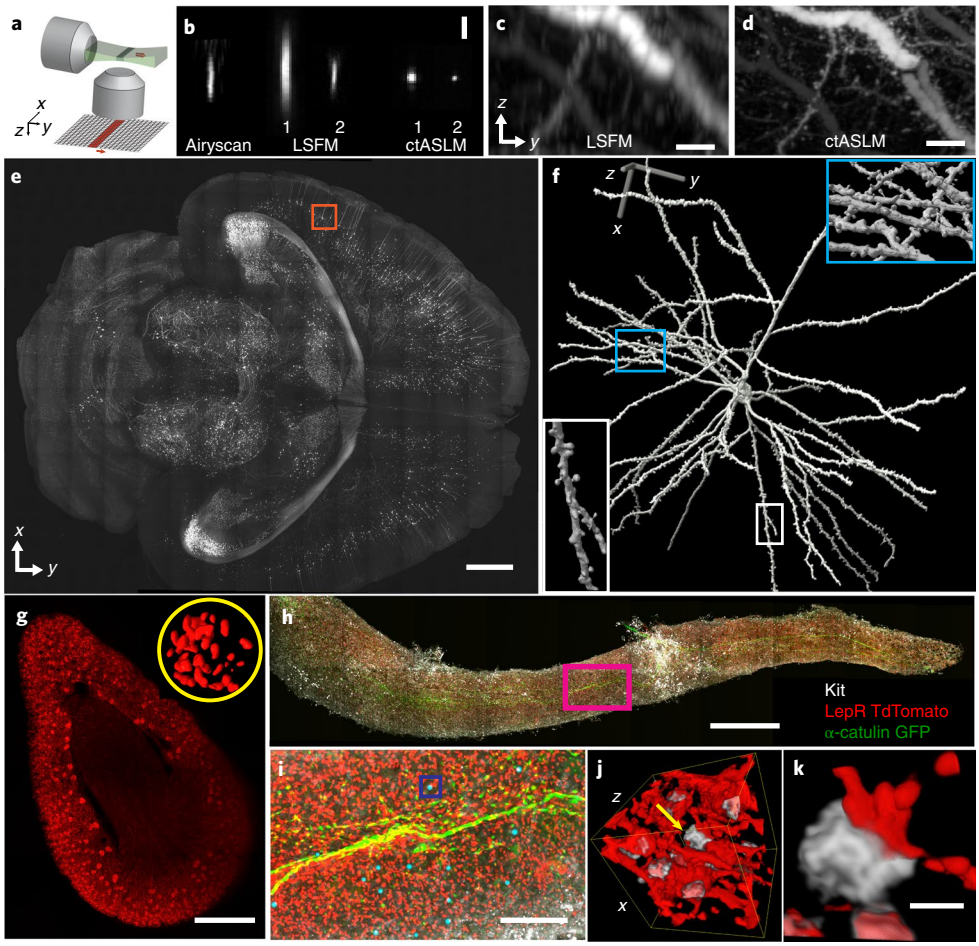

Fig. 1 | ctASLM enables isotropic submicron imaging over large fields of view. a, Working principle of ctASLM: a thin light-sheet is scanned in its propagation direction. The rolling shutter readout of an sCMOS (scientific complementary metal oxide semiconductor) camera, adjusted to the size of the beam waist, is tightly synchronized to the light-sheet scan. **b**, Experimentally measured point spread function (PSF) of Airyscan confocal microscopy, conventional LSFM and ctASLM. LSFM and ctASLM PSFs are shown for NA 0.4 (indicated by 1) and 0.7 objectives (indicated by 2). **c**,**d**, Maximum-intensity projections of CLARITY-cleared cortical Thy1-eYFP neurons as imaged by LSFM (**c**) and ctASLM (**d**). **e**, An *x*–*y* view of a section 2.5 mm from the top surface of a Thy1-eGFP PEGASOS-cleared brain. **f**, Close-up of the red box with a rendered neuron. Insets provide magnified views of synaptic spines. **g**, Volume rendering of a BABB-cleared mouse kidney labeled with Flk1-GFP. Inset shows a 3D rendering of endothelial cells in one selected glomerulus. **h**, Maximum-intensity projection of a PEGASOS-cleared mouse tibia bone marrow plug. **i**, Close-up of the magenta box shows α-catulin-GFP$^+$ cells (green, Alexa Fluor 488), lepR$^+$ niche cells (red, tdTomato) and c-Kit$^+$ cells (gray, Alexa Fluor 647). The α-catulin-GFP$^+$/c-Kit$^+$ HSCs are labeled with cyan spheres. **j**,**k**, The volume rendering of the boxed region in **h** showing the progenitor cells in gray and niche cells in red. All biological data were acquired with NA 0.4 objectives. Scale bars, 3 μm (**b**), 20 μm (**c**,**d**), 1 mm (**e**), 500 μm (**g**), 1 mm (**h**), 200 μm (**i**), 40 × 40 × 40 μm^3 (**j**) and 5 μm (**k**).

the remote focusing arm and a synchronous camera readout to deliver isotropic high-resolution, a large field of view and high-contrast imaging (Fig. 1a)[12,13]. Compared to other light-sheet variants that lack aberration-free refocusing (such as mesoSPIM) and camera synchronization (such as tiling light-sheet microscopy), ctASLM is better suited for using thin light-sheets over large fields of view (Supplementary Note 1)[14,15]. Consequently, we present two implementations of ctASLM using matched multi-immersion objectives of numerical aperture (NA) 0.4 or 0.7, which provide isotropic subcellular imaging at two resolution levels (~600 nm and ~300 nm, respectively; Supplementary Fig. 2 and Supplementary Note 2). By tiling multiple volumes[16], ctASLM permits routine visualization of subcellular features throughout millimeters of tissue, regardless of the clearing method (Supplementary Fig. 3 and Supplementary Video 1).

To evaluate optical performance, we imaged subdiffraction beads with ctASLM, conventional light-sheet microscopy and an Airyscan confocal microscope (Fig. 1b), each equipped with long working distance multi-immersion objectives (detailed full-width half-maximum values are given in Table 1). Importantly, ctASLM improves axial resolution sixfold over conventional LSFM and three to sevenfold over Airyscan confocal microscopy. Depending on the refractive index of the immersion media, ctASLM achieves an isotropic, raw resolution of 800–900 nm with the NA 0.4 objectives and 400–500 nm with the NA 0.7 objectives, over the same field of view encompassed by conventional light-sheet microscopy (Fig. 1b, Supplementary Figs. 4 and 5, Supplementary Tables 2 and 3 and Supplementary Note 3)[17,18]. Iterative deconvolution improves the resolution of ctASLM by ~30%, leading to 300 nm resolution with the NA 0.7 objective (Table 1 and Supplementary Figs. 6 and 7), which we further confirmed inside a biological sample (Supplementary Fig. 8). To the best of our knowledge, the axial resolving power of ctASLM of 260–290 nm in solvent-based media is the highest reported *z*-resolution in diffraction-limited light-sheet microscopy (without requiring super-resolution mechanisms such as stimulated emission depletion, nonlinear structured illumination or localization approaches). Owing to the excellent optical sectioning of ctASLM and high-quality raw data, deconvolution

Table 1 | Experimental values for PSFs

Microscope	NA	Field of view (μm^2)	Lateral resolution (μm)	Axial resolution (μm)	Field of view (μm^2)	Lateral resolution (μm)	Axial resolution (μm)
		($\eta = 1.333$)			($\eta = 1.56$)		
ctASLM1	0.4	870 × 870	0.95 ± 0.030 (raw, $n = 10$)	0.95 ± 0.115 (raw, $n = 10$)	737 × 737	0.83 ± 0.035 (raw, $n = 10$)	0.94 ± 0.075 (raw, $n = 10$)
		870 × 870	0.77 ± 0.033 (deconv., $n = 10$)	0.75 ± 0.087 (deconv., $n = 10$)	737 × 737	0.62 ± 0.032 (deconv., $n = 10$)	0.72 ± 0.064 (deconv., $n = 10$)
ctASLM2	0.7	327 × 327	0.48 ± 0.012 (raw, $n = 10$)	0.48 ± 0.029 (raw, $n = 10$)	310 × 310	0.45 ± 0.016 (raw, $n = 6$)	0.38 ± 0.032 (raw, $n = 6$)
		327 × 327	0.36 ± 0.012 (deconv., $n = 10$)	0.37 ± 0.018 (deconv., $n = 10$)	310 × 310	0.31 ± 0.018 (deconv., $n = 6$)	0.26 ± 0.018 (deconv., $n = 6$)
LSFM1	0.4	870 × 870	1.04 ± 0.04 (raw, $n = 5$)	6.7 ± 0.4 (raw, $n = 5$)	737 × 737	—	—
LSFM2	0.7	327 × 327	0.5 ± 0.02 (raw, $n = 5$)	2.66 ± 0.04 (raw, $n = 5$)	310 × 310	—	—
Airyscan	0.8	—	—	—	—	0.32 ± 0.03 (deconv., $n = 3$)	3.1 ± 0.4 (deconv., $n = 3$)

Note that the magnification and resolving power slightly change for the multi-immersion lenses used for ctASLM for refractive indices (η) between 1.333 and 1.56, respectively. All numbers presented are the mean ± the s.d. of the full-width half-maximum and the sample size n was 3–10 for each measurement. deconv., deconvolution; —, not measured.

is not always needed (Supplementary Fig. 9). In comparison to a conventional light-sheet microscope (Fig. 1c), individual synaptic spines could be clearly resolved in any spatial dimension (Fig. 1d) and neurons could be traced in 3D using unprocessed data.

Because of its large field of view and long working distance, we explored the ability of ctASLM equipped with the NA 0.4 objectives to image large tissues. Figure 1e shows a PEGASOS-cleared Thy1-GFP brain (Supplementary Video 2) and Fig. 1f shows a volume rendering of a neuron at a depth of 2.5 mm. Even at this depth, we were able to resolve fine neuronal structures and individual spines (Supplementary Videos 3 and 4 and Supplementary Figs. 10 and 11). Using benzyl alcohol and benzyl benzoate (BABB) clearing, we imaged a neonatal kidney labeled with the vasculature marker Flk1-GFP and maintained sufficiently high spatial resolution to distinguish individual cells throughout the $3.4 \times 2.6 \times 2.5\,mm^3$ volume (Fig. 1g and Supplementary Video 5). We also imaged the hematopoietic stem cell (HSC) niche in a PEGASOS-cleared bone marrow plug (Fig. 1h–k). HSCs are responsible for the continued production of blood and immune cells throughout life and HSCs reside within a perisinusoidal niche in the bone marrow, where leptin-receptor-positive (lepR$^+$) stromal cells and endothelial cells synthesize the factors required for HSC maintenance[19]. However, careful analysis of the interactions between these cells has not been performed because HSCs are rare (0.003% of all bone marrow cells) and imaging large specimens has been prohibitively time-consuming[19]. ctASLM imaging allowed us to identify α-catulin$^+$c-Kit$^+$ HSCs, markers that identify all HSCs in young adult bone marrow and that give very high levels of purity (Fig. 1i). The isotropic resolution revealed extensive interactions of lepR$^+$ cell projections and HSCs (Fig. 1j,k), which were distorted when imaged with a confocal microscope (Supplementary Video 6). Further, we imaged the entire PEGASOS cleared murine olfactory bulb (Supplementary Video 7), which included ossified structures that would otherwise be incompatible with aqueous clearing methods and demonstrated the importance of being able to image in solvents that range in refractive index from 1.333 to 1.559.

Next, we tested whether the quality of ctASLM data was sufficient for the application of automated and unbiased approaches to image quantification. Using the open source software u-shape3D, we could readily detect dendritic spines and cluster them on the basis of their 3D morphology (Fig. 2a and and Supplementary Videos 8 and 9)[20]. These clusters spanned a wide range of morphologies (Fig. 2b), formed a structured space following principal-component analysis (Fig. 2c) and were separable via interpretable measures, such as the ratio of spine neck area to spine surface area

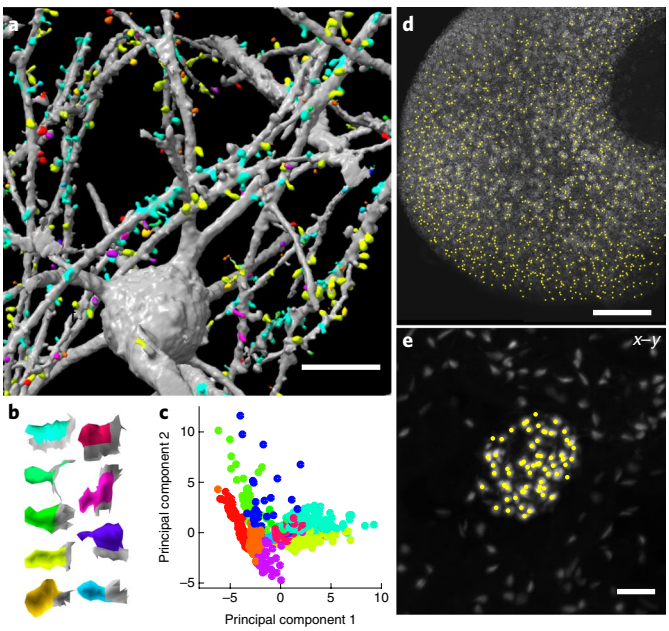

Fig. 2 | ctASLM allows detection of cellular and subcellular features in large tissues. a, Machine-learning-based supervised detection and unsupervised clustering of spines in a CLARITY-cleared mouse brain are shown. **b**, The shape of the median spine in each cluster. **c**, Principal-component analysis of the spines. **d**, Image showing the automatic detection of glomeruli in the mouse kidney dataset as shown in Fig. 1g. **e**, Lateral maximum-intensity projection of one glomerulus and the detection of individual endothelial cells (yellow circles). Data are acquired with NA 0.4 objectives. Scale bars, 20 μm (**a**), 200 μm (**d**) and 50 μm (**e**).

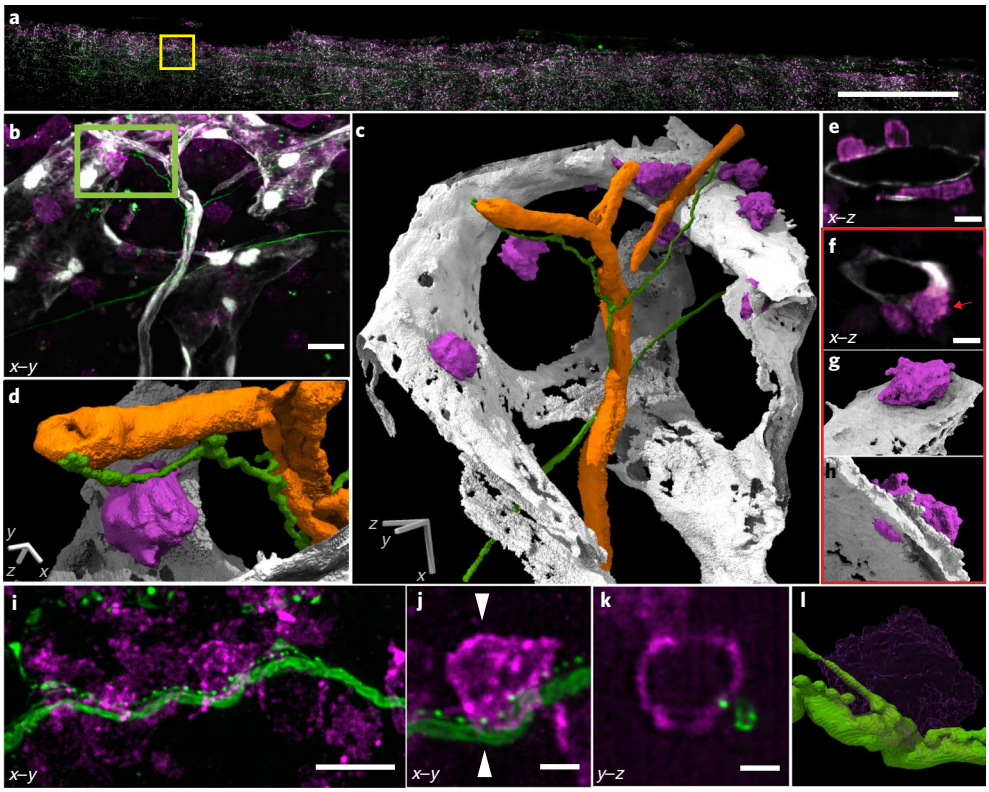

Fig. 3 | ctASLM allows visualization of subcellular interactions among cells in large tissues. a, Maximum-intensity projection of a PEGASOS-cleared mouse bone marrow specimen. **b**, Close-up of the yellow box shows c-Kit$^+$ stem/progenitor cells (magenta, Alexa 647), nerve fibers (green, Alexa 488) and blood vessels (gray, tdTomato). **c**, 3D-segmented and rendered view of **b**. Orange depicts arterioles. **d**, Close-up view (segmented and rendered) of the green box in **b** showing c-Kit$^+$ cells in contact with a nerve fiber. **e**, An *x–y* cross-section of c-Kit$^+$ cells residing in the sinusoids. **f**, c-Kit$^+$ cells are positioned along fenestrations in the walls of the sinusoids. **g**,**h**, A 3D rendering of a cell highlighted in **f**. **i**, Nerve fibers with punctate staining adjacent to c-Kit$^+$ cells in a BABB-cleared mouse bone marrow specimen. **j**, Close-up view of a possible interaction. **k**, *y–z* view along a plane indicated by white triangles in **j** showing an interaction between a nerve fiber and a c-Kit$^+$ cell. **l**, Volume rendering of **j**. Data were acquired with NA 0.7 objectives. Scale bars, 500 µm (**a**), 10 µm (**b**,**c**), 2.5 µm (**d**), 5 µm (**e**,**f**) 10 µm (**i**) and 2.5 µm (**j**,**k**).

(Supplementary Fig. 12). For the kidney, damage to glomeruli is associated with a spectrum of clinical outcomes, including proteinuria and reduced filtration, which are the hallmarks of chronic kidney diseases. To evaluate tissue composition in an automated fashion, we developed a multiscale watershed algorithm to identify glomeruli (Fig. 2d). Further, owing to the high axial resolution, we could detect endothelial cells within an individual glomerulus (Fig. 2e, Supplementary Fig. 13 and Supplementary Video 10). Formerly, such an analysis was performed manually on hundreds of serial-sectioned tissue preparations, because identification of individual cells in confocal datasets was particularly challenging owing to its diminished *z*-resolution.

Lastly, we imaged PEGASOS-cleared HSCs, nerve fibers and vasculature in mouse bone marrow with the NA 0.7 objectives and 300-nm-scale resolution (Fig. 3a–d and Supplementary Fig. 14). Many c-Kit$^+$ HSCs were associated with fenestrations in the sinusoidal blood vessels, consistent with observations that HSCs migrate through these fenestrations to enter and exit circulation (Fig. 3e–h)[19]. Importantly, detailed features of the HSCs could not be resolved with a confocal microscope (Supplementary Fig. 15), but were readily resolved by ctASLM, owing to its improved resolution. Intriguingly, in BABB-cleared bone marrow plugs we observed punctate staining along nerve fibers that resembled synapses, raising the possibility that nerve fibers directly interact with a subset of c-Kit$^+$ cells (Fig. 3i–l and Supplementary Video 11).

In summary, ctASLM simultaneously provides subcellular detail and tissue-scale anatomy. Because tissues are inherently 3D, the isotropic resolution of ctASLM is an important attribute to describe 3D morphologies and molecular concentrations in an unbiased way. Indeed, even higher resolutions can be achieved by combining ctASLM with expansion microscopy (Supplementary Figs. 16–18 and Supplementary Videos 12 and 13), albeit with one order of magnitude larger field of view than lattice light-sheet microscopy. Of note, ctASLM imposes a higher out-of-focus excitation load on the sample than a conventional light-sheet system and illuminates a smaller portion at a time, both of which can increase photobleaching. However, even at the highest resolution level (NA 0.7), we were not limited by photobleaching and were able to image a sample multiple times with a high signal-to-noise ratio. Some steering of the light-sheet was observed deep in the PEGASOS brain and likely higher-order aberrations were present in some of our CLARITY-cleared samples (Supplementary Fig. 19). Automated alignment routines, as well as adaptive optics, could help to achieve diffraction-limited performance in less than optimally cleared samples. With the given resolution level, we readily show that we can combine ctASLM with computer vision to detect and classify biological features throughout millimeters of tissue. Consequently, we believe that ctASLM will expedite human cell atlas efforts, providing much-needed insight into how tissue function manifests in health and disease, from the heterogenous cellular populations that compose it.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information,

acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41592-019-0615-4.

Received: 10 April 2019; Accepted: 17 September 2019;
Published online: 31 October 2019

References

1. Villani, A.-C. et al. Single-cell RNA-seq reveals new types of human blood dendritic cells, monocytes and progenitors. *Science* **356**, eaah4573 (2017).
2. Wang, K. et al. Rapid adaptive optical recovery of optimal resolution over large volumes. *Nat. Methods* **11**, 625–628 (2014).
3. Richardson, D. S. & Lichtman, J. W. Clarifying tissue clearing. *Cell* **162**, 246–257 (2015).
4. Welf, ErikS. et al. Quantitative multiscale cell imaging in controlled 3D microenvironments. *Dev. Cell* **36**, 462–475 (2016).
5. Huisken, J. Optical sectioning deep inside live embryos by selective plane illumination microscopy. *Science* **305**, 1007–1009 (2004).
6. Dodt, H.-U. et al. Ultramicroscopy: three-dimensional visualization of neuronal networks in the whole mouse brain. *Nat. Methods* **4**, 331–336 (2007).
7. Stefaniuk, M. et al. Light-sheet microscopy imaging of a whole cleared rat brain with Thy1–GFP transgene. *Sci. Rep.* **6**, 28209 (2016).
8. Pende, M. et al. High-resolution ultramicroscopy of the developing and adult nervous system in optically cleared *Drosophila melanogaster*. *Nat. Commun.* **9**, 4731 (2018).
9. Tomer, R. et al. SPED light sheet microscopy: fast mapping of biological system structure and function. *Cell* **163**, 1796–1806 (2015).
10. Buytaert, J. A. & Dirckx, J. J. Tomographic imaging of macroscopic biomedical objects in high resolution and three dimensions using orthogonal-plane fluorescence optical sectioning. *Appl. Opt.* **48**, 941–948 (2009).
11. Santi, P. A. et al. Thin-sheet laser imaging microscopy for optical sectioning of thick tissues. *Biotechniques* **46**, 287–294 (2009).
12. Dean, KevinM. et al. Deconvolution-free subcellular imaging with axially swept light sheet microscopy. *Biophysical J.* **108**, 2807–2815 (2015).
13. Botcherby, E. J. et al. Aberration-free three-dimensional multiphoton imaging of neuronal activity at kHz rates. *Proc. Natl Acad. Sci. USA* **109**, 2919–2924 (2012).
14. Gao, L. Extend the field of view of selective plan illumination microscopy by tiling the excitation light sheet. *Opt. Express.* **23**, 6102–6111 (2015).
15. Voigt, F. F. et al. The mesoSPIM initiative: open-source light-sheet microscopes for imaging cleared tissue. *Nat. Methods* https://doi.org/10.1038/s41592-019-0554-0 (2019).
16. Hörl, D. et al. BigStitcher: reconstructing high-resolution image datasets of cleared and expanded samples. *Nat. Methods* **16**, 870–874 (2019).
17. Gao, R. et al. Cortical column and whole-brain imaging with molecular contrast and nanoscale resolution. *Science* **363**, eaau8302 (2019).
18. Fu, Q., Martin, B. L., Matus, D. Q. & Gao, L. Imaging multicellular specimens with real-time optimized tiling light-sheet selective plane illumination microscopy. *Nat. Commun.* **7**, 11088 (2016).
19. Crane, G. M., Jeffery, E. & Morrison, S. J. Adult haematopoietic stem cell niches. *Nat. Rev. Immunol.* **17**, 573–590 (2017).
20. Driscoll, M. K. et al. Robust and automated detection of subcellular morphological motifs in 3D microscopy images. *Nat. Methods* **16**, 1037–1044 (2019).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2019

Methods

Animal specimens. All animal protocols were approved by local Institutional Animal Care and Use Committees (IACUC) as directed by the National Institutes of Health, and strictly followed. These protocols include AC-AAAR0417 (to R.T., Columbia University), 2017-102370 (to J.M., UT Southwestern Medical Center), 101917 (to D.M., UT Southwestern Medical Center), 101715 (to I.B., UT Southwestern Medical Center) and 102632 (to S.M., UT Southwestern Medical Center).

Microscope control. The data acquisition computer for ctASLM v.1 was a Dell Precision 5810 tower equipped with an Intel Xeon E5-1650W v.3 processor operating at 3.5 GHz with six cores and 12 threads, 128 GB of 2,133 MHz DDR4 RAM and an integrated Intel AHCI chipset controlling 4 × 512 GB SSDs in a RAID0 configuration. The data acquisition computer for ctASLM v.2 was a Colfax ProEdge SXT9800 workstation equipped with two Intel Xeon Silver 4112 processors operating at 2.6 GHz with eight cores and 16 threads, 96 GB of 2.667 GHz DDR4 RAM, a Intel DC P3100 1024 GB M.2 NVMe drive and a Micron 5200 ECO 7,680 GB hard drive for file storage. All software was developed using a 64-bit version of LabView 2016 equipped with the LabView Run-Time Engine, Vision Development Module, Vision Run-Time Module and all appropriate device drivers, including NI-RIO drivers (National Instruments). Software communicated with the camera (Flash 4.0, Hamamatsu) via the DCAM-API for the Active Silicon Firebird frame-grabber and delivered a series of deterministic transistor–transistor logic (TTL) triggers with a field programmable gate array (PCIe 7852R, National Instruments). These triggers included control of the resonant mirror galvanometers, voice coils, stage positioning, laser modulation and blanking, camera fire and external trigger. Some of the core functions and routines in the microscope control software are licensed under a Material Transfer Agreement from Howard Hughes Medical Institute, Janelia Research Campus. The control software code for ctASLM can be requested from the corresponding authors and will be distributed under a material transfer agreement with Howard Hughes Medical Institute, Janelia Research Campus.

ctASLM v.1 microscope layout. A schematic layout of the ctASLM v.1 microscope is shown in Supplementary Fig. 20 and a parts list is provided in Supplementary Table 4. The illumination train consisted of four Coherent Obis lasers (LX 405-100C, LX 488-50C, LS 561-50 and LX 637-140C) that were combined with dichroic beam splitters (LM01-427-25, LM01-503-25 and LM01-613-25, Semrock), focused through a 30-µm pinhole (P30D, ThorLabs), with a 50-mm achromatic doublet (AC254-50-A, ThorLabs), recollimated with a 200-mm achromatic doublet, (AC254-200-A-ML, ThorLabs), and directed through a 3× Galilean beam expander (GEB03-A). Thus, the initial beams were expanded by a factor of 12 before being focused onto a resonant mirror galvanometer (CRS 4 kHz, Cambridge Technology) with a cylindrical lens (ACY254-50-A, ThorLabs). The mirror galvanometer was used to wobble the light sheet and was driven with a 12 volt power supply (A12MT400, Acopian), and the one-dimensional focus of the cylindrical lens was relayed with a 200-mm achromatic doublet (AC508-200-A, ThorLabs), passed through a 50:50 polarizing beam splitter and a quarter waveplate and imaged onto a mirror with a ×4 microscope objective (XL Fluor ×4, NA 0.28, Olympus Life Sciences). The mirror was mounted on a voice coil with 10 mm of travel, sub-50-nm positioning repeatability and a sub-3-ms response time (LFA-2010, Equipment Solutions). The reflected light was recollected by the same ×4 objective, the polarization state was rotated by the second passage through the quarter waveplate (AQWP3, Boldervision) reflected with the cube polarizing beam splitter (10FC16PB.7, Newport) and relayed to the illumination objective (Cleared Tissue Objective, NA 0.4 Advanced Scientific Imaging) with 200-mm and 75-mm achromatic doublets (AC508-200-A and AC508-75-A, ThorLabs). To correct for spherical aberrations, a 3-mm N-BK7 glass piece (37–005, Edmund Optics) was placed between the ×4 objective and the voice coil actuated mirror (Supplementary Fig. 21).

The detection arm consisted of an identical microscope objective (Cleared Tissue Objective, Advanced Scientific Imaging), a tube lens (ITL200-A, ThorLabs), a 4-position filter wheel (LB10-WHS4 and a Lambda 10-3, Sutter Instruments) and a sCMOS camera (Flash 4.0, Hamamatsu Corporation). For emission filters, we used 445/20, 525/30, 605/15 and 647 long-pass filters (FF01-444/20-25, FF01-525/30-25, FF01-505/15-25 and BLP01-647R-25, Semrock), for blue, green, red and far-red, respectively. Both the illumination and detection objectives were immersed in a large chamber that was used to house the specimen. To collect a *z*-stack, a voice coil direct drive was used to scan the specimen in a sample-scanning mode (V-522.1AA, C-413.2GA and C-413.1IO, Physik Instrumente). Subvolumes were acquired by moving the position of the specimen and the voice coil stage with a three-axis motorized stage (3DMS and MP-285A, Sutter Instruments). Imaging a 1-mm^3 volume using ctASLM took 17.6 min, using a 100-ms exposure time, an isotropic step size of 0.425 µm and 18% overlap between stacks for faithful stitching.

ctASLM v.2 microscope layout. ctASLM v.2 was built essentially identical to ctASLM v.1, with a few minor differences. A schematic drawing of the setup is provided in Supplementary Fig. 22 and a parts list is provided in Supplementary Table 5. The illumination train consisted of three Coherent Obis lasers (LS 561-100C, LX 647-120C and 488-150C) housed in an Obis LX/LS Laser Box that permitted user-selectable input impedances and improved thermal management. All three lasers were combined with dichroic beam splitters (LM01-503-25 and LM01-613-25, Semrock), their polarization rotated with an achromatic half waveplate (10RP52-1B, Newport), focused with a 50-mm achromatic doublet (AC254-50-A, ThorLabs) through a 30-µm pinhole (P30D, ThorLabs), recollimated with a 100-mm achromatic doublet (AC254-100-A, ThorLabs) and expanded an additional five times with a beam expander (BEO5M, ThorLabs). A cylindrical lens (ACY254-50-A, ThorLabs) was used to create a one-dimensional laser-line focus on a 4-kHz resonant galvo (6SC04KA040-01Y, Cambridge Technology) that was used to wobble the light sheet and was driven with a 12 volt power supply (A12MT400, Acopian). The one-dimensional focus of the cylindrical lens was relayed with a 100-mm achromatic doublet (AC254-100-A, ThorLabs), passed through a polarizing beam splitter (10FC16PB.7, Newport), an achromatic quarter waveplate (10RP54-1B, Newport) and imaged onto a mirror with a ×40 microscope objective (CFI S Plan Fluor ELWD ×40, NA 0.60, Nikon Instruments). The mirror was mounted on a voice coil with 10 mm of travel, sub-50-nm positioning repeatability and a sub-3-ms response time (LFA-2010, Equipment Solutions). The reflected light was recollected by the same ×40 objective, the polarization state was rotated by the second passage through the quarter waveplate, reflected with the polarizing beam splitter and relayed to the illumination objective (Cleared Tissue Objective, NA 0.7 Advanced Scientific Imaging) with 80-mm and 200-mm achromatic doublets (AC508-200-A and AC508-75-A, ThorLabs).

The detection arm consisted of an identical NA 0.7 microscope objective (Cleared Tissue Objective, Advanced Scientific Imaging), a 300-mm achromatic doublet (AC500-300-A, ThorLabs), a four-position filter wheel (LB10-WHS4 and a Lambda 10-3, Sutter Instruments) and a liquid cooled (ThermoCube 400, Solid State Cooling Systems) sCMOS camera (Flash 4.0, Hamamatsu Corporation). For emission filters, we used 525/30, 605/15 and 647 long-pass filters (FF01-525/30-25, FF01-505/15-25 and BLP01-647R-25, Semrock), for green, red and far-red, respectively. Both the illumination and detection objectives were immersed in a large chamber that was used to house the specimen. To collect a *z*-stack, a 480-µm travel piezo (Piezo Jena NanoX 400) was used to scan the specimen in a sample-scanning mode. Subvolumes were acquired by moving the position of the specimen and the piezo stage with a three-axis motorized stage (3DMS and MP-285A, Sutter Instruments).

Sample mounting. Samples were glued with cyanoacrylate (PT09, Pacer Technology) or with silicone (GE Silicone 2+) onto a 1 × 2.5 cm^2 glass slide cut from a 2.5 × 7.5 cm^2 microscope slide (catalog no. 3050, Thermo Scientific). The glass slides were mounted with a custom holder to the sample-scanning stage. Computer-aided design drawings of the sample holder assembly are shown in Supplementary Fig. 23.

Conventional light-sheet microscopy. For conventional low-NA light-sheet imaging, the same optical train was used as in ctASLM v.1, but with the presence of a variable width slit aperture conjugate to the back pupil of the cylindrical lens and without axial scanning of the illumination beam. The illumination NA was selected so that the entire field of view (870 × 870 μm^2) resided within two Rayleigh lengths of the illumination beam. This resulted in a Gaussian beam with a 10^{-2} thickness of ~7 µm, which is in agreement with resolution measurements obtained using subdiffraction fluorescence beads. With the ctASLM v.2 microscope (NA 0.7 objectives), an axial resolution of 2.5 µm resulted, which we believe is in large part owing to the close to diffraction-limited detection PSF.

Confocal microscopy. Imaging of the bone plug was performed on a Zeiss LSM880 confocal microscope equipped with a long working distance multi-immersion objective (LD LCI Plan-Apo ×25 0.8 NA) as previously described[21,22]. PEGASOS-cleared mouse brain was also imaged using a Leica SP8 confocal laser scanning microscope equipped with an HCX APO ×20/0.95 immersion objective. For the best performance a 56-µm pinhole size with a pixel dwell time of 1.84 µs was used. A comparison of ctASLM to the data acquired with the Leica SP8 is shown in Supplementary Fig. 24.

Sample preparation. *Zebrafish.* The zebrafish embryo was a transgenic line (krdl:mCherry) expressing mCherry on the vascular endothelium. A 3-d post-fertilization zebrafish embryo was mounted in 2% low-melting agarose (A9045-25G, Sigma-Aldrich) in a fluorinated ethylene propylene (FEP) tube (FEP HS 0.029 EXP/0.018 REC, Zeus). The working solvent of the agarose was E3 medium. The FEP tube was sonicated in 70% ethanol and then washed with clean distilled water before mounting the zebrafish embryo. We first selected and left a zebrafish embryo in ~50 µl of E3 medium on a Petri dish. Next, 5 µl of 0.1% tricaine was added to anesthetize the embryo. Approximately 50 µl of the mixture of the tricaine and E3 medium was removed and liquid low-melting agarose was added to the embryo. The embryo in the liquid agarose was aspirated directly into the clean FEP tube with a pipette. Within a few minutes the agarose was solidified and the zebrafish embryo was mounted in the FEP tube. For the last step, we used a needle to make two holes in the FEP tube above and below the embryo to increase gaseous

exchange, allowing the embryo to breathe. The embryo was then transferred into the microscope chamber and was ready to be imaged.

CLARITY brain. Thy1-eYFP brain was extracted and cleared using passive CLARITY[23] with the following hydrogel monomer (HM) solution recipe: 1% (wt/vol) acrylamide, 0.05% (wt/vol) bisacrylamide, 4% paraformaldehyde, 1× PBS, deionized water and 0.25% thermal initiation VA-044 (Wako Chemicals, NC0632395). Transcardiac perfusion was performed with 20 ml of HM solution, followed by brain extraction and overnight incubation at 4 °C. Hydrogel was polymerized by incubating the brain samples at 37 °C for 3–4 h and clearing was performed by shaking in clearing buffer (4% (wt/vol) SDS and 0.2 M boric acid (pH 8.5)) at 37 °C for ~3 weeks. After washing with the buffered solution (0.2 M boric acid buffer (pH 7.5) and 0.1% Triton X-100), brains were transferred into 50–87% glycerol solution for refractive index matching (as described previously[24]) and imaging.

Olfactory bulb. Double transgenic mice were made by crossing Ai9 mice (B6.Cg-Gt(ROSA)26Sortm9(CAG-tdTomato)Hze/J, Jax Laboratory catalog no. 007909) with FosTRAP mice[25] and the resulting mice were termed cFosxAi9 mice. Briefly, animals were anesthetized with intraperitoneal injections of ketamine/xylazine (120 mg kg^{-1} ketamine and 16 mg kg^{-1} xylazine). After verification of complete anesthesia via the toe pinch test, the chest cavity was exposed and a syringe needle was inserted into the left ventricle. Mice were perfused with 20 ml of PBS solution followed by 20 ml of a 4% PFA solution. After perfusion, mice were decapitated and the brain, including the olfactory bulbs, were carefully dissected out and placed into 4% PFA solution overnight. These samples were then cleared using the PEGASOS method as described by Jing et al.[26]. All samples were kept at 42 °C for this process. Briefly, samples were decolorized for 2 d in 25% Quadrol before undergoing delipidation in a series of 30%, 50% and 75% tert-butanol solutions. This was followed by a final delipidation step with TBP (75% tert-butanol and 25% PEGMMA solution). Finally, samples were submerged in BB-PEG (75% benzyl benzoate, 22% PEGMMA and 3% Quadrol) until imaging. The data shown here were on the basis of a male mouse aged 3.32 months.

Bone marrow. For imaging of blood vessels or niche cells, double or triple transgenic mice were generated by crossing B6.Cg-Gt(ROSA)26Sortm9(CAG-tdTomato)Hze/J (Jax no. 007909) mice to Cdh5-creER (B6.CBA-Tg(Cdh5-cre/ERT2)1Rha) or leptin receptor (lepR)-cre and α-catulin-GFP mice[21,27]. To induce CreER recombinase activity, Cdh5-creER;Rosa26-tdTomato mice received an intraperitoneal injection of 2 mg of tamoxifen dissolved in corn oil every second day for 10 d (five injections in total). For imaging of blood vessels, bones from Cdh5-creER;Rosa26-tdTomato mice were embedded into OCT embedding medium. Using a cryostat, femurs were then cut in half lengthwise and then re-mounted at a 90 degree angle to the original cut and cut in half lengthwise again, resulting in a quarter bone. For all other bone marrow samples, intact bone marrow plugs from freshly dissected tibias of 6–8-week-old mice were extruded from the bone using a PFA-filled syringe with a 25-gauge needle and placed directly into 4% PFA solution for 6 h at room temperature. Processing of samples was carried out as previously described by Acar et al.[21]. Briefly, fixed quarter bones or plugs were washed in PBS and blocked in whole-mount staining medium (PBS with 5% donkey serum, 0.5% NP-40 and 10% DMSO) overnight at room temperature. After blocking, samples were stained for 3 d at room temperature with primary antibodies (lepR-cre;Rosa26-tdTomato;α-catulin-GFP sample: chicken anti-GFP: GFP-1020, Aves Labs and goat anti-c-Kit: BAF1356, R&D Systems) or (Cdh5-creER;Rosa26-tdTomato sample and C57Bl/6 nerve/c-Kit stained sample: rabbit anti-tyrosine hydroxylase at 1:250 (EMD Millipore catalog no. AB152), rabbit anti-peripherin at 1:250 (EMD Millipore catalog no. AB1530) and goat anti-c-Kit at 1:250 (R&D Systems catalog no. BAF1356)) in staining solution. Then the samples were washed multiple times in PBS at room temperature for 1 d and put into staining solution containing secondary antibodies (lepR-cre;Rosa26-tdTomato;α-catulin-GFP sample: Alexa Fluor 488-AffiniPure F(ab′)2 fragment donkey anti-chicken IgY and Alexa Fluor 647-AffiniPure F(ab′)2 fragment donkey anti-goat IgG) or (Cdh5-creER;Rosa26-tdTomato sample and C57Bl/6 nerve/c-Kit stained sample: Alexa Fluor 488-AffiniPure F(ab′)2 fragment donkey anti-rabbit IgG, Alexa Fluor 647-AffiniPure F(ab′)2 fragment donkey anti-goat IgG) for 3 d followed by a 1-d wash to remove any unbound secondary antibodies. C57Bl/6 nerve/c-Kit stained marrow plugs (Fig. 3i–k) were embedded in 2% low-melt agarose in water and were dehydrated in a series methanol washes, followed by refractive index matching for 1 h at room temperature in 1:2 BABB. Cdh5-creER;Rosa26-tdTomato quarter bones (Fig. 3a–h) and lepR-cre;Rosa26-tdTomato;α-catulin-GFP bone marrow niche plugs embedded in 2% low-melt agarose (Fig. 1h–k) were prepared using the PEGASOS method[26]. Briefly, samples were dehydrated in a series of 30%, 50%, 70% and 100% tert-butanol. Each step was performed for 8 h at room temperature. Samples were then refractive index-matched overnight at room temperature in BB-PEG. BB-PEG was prepared as a 3:1 mixture of BB (Sigma-Aldrich B6630) and PEGMMA500 (Sigma-Aldrich 447943) supplemented with 3% w/v Quadrol (Sigma-Aldrich 122262). Samples were imaged from 1 d to 2 months after initial refractive index matching without marked loss of endogenous tdTomato or Alexa Fluor signal.

Isolated hippocampus. The Thy1-GFP mouse was transcardially perfused with cold PBS, then 4% PFA in PBS for fixation. The brain was taken out and put into 4% PFA in PBS overnight at 4 °C, then 2-mm thick hippocampal slices were cut coronally with a vibratome. The hippocampal slices were then cleared with PEGASOS-clearing protocol, previously described by Jing et al.[26]. Briefly, the slices were decolorized in 25% Quadrol for 1 d, then delipidated in series of 30%, 50% 70% tert-butanol solutions, dehydrated in tB-PEG and finally cleared in BB-PEG.

Expanded hippocampus. For imaging, expanded hippocampal samples from either Thy1-GFP mice (Supplementary Figs. 16–18) or wild-type mice injected with Synaptotag AAV (Supplementary Fig. 17) were used[28]. The mice were transcardially perfused with cold PBS, then 4% PFA in PBS for fixation. The brain was taken out and put into 4% PFA in PBS overnight at 4 °C, then 0.1-mm thick coronal hippocampal slices were cut with a vibratome. Samples were processed according to the original protein-expansion microscopy procedure[17,29]. Briefly, slices were permeabilized using 5% BSA, 0.5% Triton X-100 in PBS for 1 h at room temperature and stained with rat anti-mCherry (Thermo Scientific M11217, 1:500; Supplementary Fig. 17) or chicken anti-GFP (Abcam ab13970, 1:500; Supplementary Fig. 18) and rabbit anti-Homer1 (Synaptic systems, 160003, 1:200) antibodies (Supplementary Fig. 18) in permeabilization buffer for 2 d at 4 °C. For Homer1 staining, antigen was retrieved by incubation in 1 mM EDTA at 60 °C for 45 mins. Slices were washed three times in PBS and stained with donkey anti-rat-Alexa594 (Invitrogen, 1:200; Supplementary Fig. 17) or goat anti-chicken-Alexa 488 (Invitrogen, 1:200) and goat anti-rabbit-Atto647N (Sigma-Aldrich, 1:200; Supplementary Fig. 18) conjugated antibodies at room temperature for 4 h. After three washing steps, samples were incubated overnight with a coupling reagent 0.1% acryoyl-X succinimidyl ester in PBS[29]. For results shown in Supplementary Fig. 16, the samples were directly crosslinked with acryoyl-X succinimidyl ester reagent without further antibody staining against GFP. Slices were washed three times with PBS for 5 min and incubated in polymerization solution (8.6% sodium acrylate, 2.5%, 0.15% *N*,*N*′-methylenebisacrylamide, 2 M NaCl, 0.2% APS, 0.2% TEMED and 0.01% TEMPO) for 1 h at 4 °C before polymerization at 37 °C for 2 h. Embedded tissue was transferred to isotonic proteinase K buffer (50 mM Tris-HCl (pH 8.0), 2 M NaCl, 1 mM EDTA, 0.5% Triton X-100 and 8 U ml^{-1} proK) and incubated for 2 h at room temperature. Complete proK digestion was achieved overnight at room temperature in proteinase K buffer (50 mM Tris-HCl (pH 8.0), 1 M NaCl, 1 mM EDTA, 0.5% Triton X-100 and 8 U ml^{-1} proK). The next day, samples were expanded for 1 h in double-distilled H_2O with three water changes. Glass holders were covered with 0.1% poly-L-lysine for 1 h, washed three times with double-distilled H_2O and allowed to air dry for 1 h. The hippocampus regions were excised from coronary slices and mounted on poly-L-lysine-coated glass holders using wax as additional support on the sides. For long-term storage and image acquisition, samples were transferred to 1 mM NaOH buffered solution.

Neonatal kidney preparation. On postnatal day 3, kidneys from an Flk1-GFP (Jackson Laboratories, 017006) mouse were harvested and fixed for 2 h at room temperature in 4% paraformaldehyde in PBS. Procedures were performed according to University of Texas Southwestern Medical Center IACUC-approved guidelines. Fixed kidneys were incubated with tert-butanol/water mixtures of increasing alcohol concentrations (30%, 50%, 70%, 80%, 96% and twice 100%) with 2% triethanolamine. The kidneys were left in BABB in a 1:2 volume ratio with 10% triethanolamine until the tissues cleared[6]. For prolonged storage in clearing medium (>2 d), samples were stored at 4 °C. Given that the tissue was harvested from neonatal mice, and no genotyping was performed, the sex of the mouse was unknown.

Stitching and 3D visualization. Image processing was performed on a Windows 10-based workstation equipped with two Intel Xeon Gold 5120 CPUs, 1 TB of RAM and an NVIDIA Quadro P6000 GPU. To stitch the subvolumes, the Fiji-based plugin BigStitcher[30] was used. Image analysis was performed with Fiji[31] and MATLAB (Mathworks) and 3D renderings were produced with ChimeraX or Arivis.

Deconvolution. We tested the Richardson–Lucy deconvolution and blind deconvolution (Supplementary Table 6 lists details for each dataset shown in this manuscript). When using the Richardson–Lucy algorithm to deconvolve the images, the PSF was extracted from the image of a single bead, centered and background-subtracted and 3D deconvolution was performed with 20 iterations of the Richardson–Lucy algorithm (Microvolution). For the blind deconvolution, we used the deconvblind function in MATLAB (Mathworks) and the following procedure. Before the deconvolution, we rotationally averaged, along the optical axis, the experimental image of a single 200-nm bead. We measured the mean value of the background in the rotationally averaged PSF and subtracted it from the PSF image. To improve this experimental PSF, we first deconvolved the raw image with the rotationally averaged and background-subtracted PSF for five iterations. This yielded a restored PSF from the devonvblind routine (the resulting PSF more closely approximated the real PSF in the measurement). In a second step, we used this restored PSF to perform blind deconvolution on the raw image data, which then resulted in the final deconvolved image. For biological data we

used 5–10 iterations, whereas for the bead analysis, we limited the number of iterations to five. The raw images and the experimental PSF were resampled twice with the bicubic algorithm with ImageJ before the deconvolution, which was particularly necessary for the ctASLM v.1 data (NA 0.4), which did not have much Nyquist oversampling in its data acquisition. ctASLM v.2 was designed with more oversampling (a larger optical magnification), hence computational resampling before deconvolution was not necessary.

Data analysis. *HSC detection.* HSCs were detected as previously described[21]. Briefly, individual α-catulin-GF$^+$ and c-Kit$^+$ HSCs were identified manually using the orthoslicer function of Bitplane Imaris v.9.2.1 software. HSC coordinates and size were interactively annotated using the Imaris spots function in the manual mode. c-Kit and lepR tdTomato surfaces were generated in Imaris using the isosurface function on the basis of an empirically chosen intensity threshold.

Glomeruli detection. In an effort to automatically detect the abundance of glomeruli throughout the entire neonatal kidney tissue, subvolumes (1,024×1,024×470 pixels) were retrieved from the stitched volume (3,042×4,018×2,757 pixels) and loaded into MATLAB. Subvolumes were subjected to a 3D Gaussian blur of 20 pixels before individual glomeruli were detected using an autoadaptive watershed algorithm implemented in MATLAB[32]. This algorithm uses a difference of Gaussians to suppress features that are either too large or small, here defined as 100 and 10 pixels, respectively. Segmented objects touching the border of the subvolume were removed. After identification of glomeruli, the encompassing region was again loaded into MATLAB and the full resolution was subjected to the same watershed algorithm, but with a difference of Gaussians of 10 and 2 pixels, respectively.

Spine detection and clustering. We detected spines using previously described software[20]. Images were first deconvolved for ten iterations using the MATLAB function deconvblind(), which uses a maximum likelihood algorithm to infer the fluorescence distribution in the sample. Following a gamma transform of 0.4 and 3D hole filling, a triangle mesh representing the cell surface was created as an isosurface at the Otsu threshold value of the image. The mesh was then decomposed into approximately convex patches using the parameters previously described, except that merging via the triangle criterion was disabled and merging via the line-of-sight criterion was performed more conservatively with a parameter of 0.8. Training data for patch merging and patch classification machine-learning models were then generated from nine non-overlapping images of size 200 pixels in the lateral dimensions and approximately 150 pixels in the axial dimension. These nine images were cropped from eight larger images. Using leave-one-out cross validation on patches labeled as certainly spines or certainly not spines, we calculated that our spine detection workflow had a precision of 0.86 and a recall of 0.88.

We performed an unsupervised clustering of detected spines into 15 clusters using agglomerative hierarchical clustering. For each cluster, we calculated the median of each of the morphological feature vectors used for clustering and plotted the spine closest to that median (Fig. 2b). Only the clusters with a median spine attached to the neuron body were plotted. Principal-component analysis was performed on these same features. We defined the spine neck surface area as the patch closure surface area. We previously defined the closure surface area as the additional surface area needed to minimally generate a closed polyhedron from the open polyhedron comprising the dendritic spine. We also defined the variation from a sphere by fitting each spine to a sphere and measuring the ratio of the standard deviation of distances from the sphere at each mesh face to the sphere radius.

Segmentation of neurons. The mouse brain volume was tiled in subvolumes for parallel processing and the signal was enhanced using a volumetric Laplacian-of-Gaussian transform with an isotropic scale of 1.5 voxels. The resulting volume was then transformed into a triangular mesh using an empirically chosen threshold. To reduce clutter, only the connected components with a volume superior to the 99 percentiles of all connected component volume are displayed in Fig. 1f. The algorithm was implemented using the open source G'MIC software (https://gmic.eu) and the mesh was imported into Meshlab[33] for coloring and image capture.

Statistics and reproducibility. For statistical analysis, we reported the mean, s.d. and number of observations. For PSF measurements of the different microscopes and immersion media, the number of measurements ranged from 5–10 for light-sheet microscopy and three for Airyscan microscopy. The values are listed in Table 1.

For repeatability, generally, only one sample was used for each imaging experiment (imaging of whole mouse brains and kidneys), unless otherwise noted. Two expanded brain slices were prepared and imaged with either the NA 0.4 or NA 0.7 version of ctASLM. The number of times that each imaging operation was repeated for each figure is indicated in Supplementary Table 6 and was one in most cases. Segmentation of neurons was performed on one dataset from a CLARITY brain subvolume and on one dataset from a hippocampal slice subvolume that was cleared with PEGASOS. Synaptic spines were detected on two different subvolumes from a CLARITY brain.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The datasets acquired for this study are available from the corresponding authors upon reasonable request.

Code availability

The instrument control software can be requested for academic use from the corresponding authors and will be delivered under material transfer agreements with Howard Hughes Medical Institute and UT Southwestern Medical Center.

References

21. Acar, M. et al. Deep imaging of bone marrow shows non-dividing stem cells are mainly perisinusoidal. *Nature* **526**, 126–130 (2015).
22. Comazzetto, S. et al. Restricted hematopoietic progenitors and erythropoiesis require scf from leptin receptor + niche cells in the bone marrow. *Cell Stem Cell* **24**, 477–486 (2018).
23. Tomer, R., Ye, L., Hsueh, B. & Deisseroth, K. Advanced CLARITY for rapid and high-resolution imaging of intact tissues. *Nat. Protoc.* **9**, 1682–1697 (2014).
24. Migliori, B. et al. Light-sheet theta microscopy for rapid high-resolution imaging of large biological samples. *BMC Biol.* **16**, 57 (2018).
25. Guenthner, CaseyJ. et al. Permanent genetic access to transiently active neurons via TRAP: targeted recombination in active populations. *Neuron* **78**, 773–784 (2013).
26. Jing, D. et al. Tissue clearing of both hard and soft tissue organs with the PEGASOS method. *Cell Res.* **28**, 803–818 (2018).
27. Sorensen, I., Adams, R. H. & Gossler, A. DLL1-mediated notch activation regulates endothelial identity in mouse fetal arteries. *Blood* **113**, 5680–5688 (2009).
28. Xu, W. & Sudhof, T. C. A neural circuit for memory specificity and generalization. *Science* **339**, 1290–1295 (2013).
29. Tillberg, P. W. et al. Protein-retention expansion microscopy of cells and tissues labeled using standard fluorescent proteins and antibodies. *Nat. Biotechnol.* **34**, 987–992 (2016).
30. Preibisch, S., Saalfeld, S. & Tomancak, P. Globally optimal stitching of tiled 3D microscopic image acquisitions. *Bioinformatics* **25**, 1463–1465 (2009).
31. Schindelin, J. et al. Fiji: an open source platform for biological image analysis. *Nat. Methods* **9**, 676–682 (2012).
32. Applegate, K. T. et al. plusTipTracker: quantitative image analysis software for the measurement of microtubule dynamics. *J. Struct. Biol.* **176**, 168–184 (2011).
33. Cignoni, P. et al. MeshLab: an open source mesh processing tool. In *Proc. Eurographics Italian Chapter Conference* (eds Scarano, V. et al) 129–136 (Eurographics Association, 2008).

Acknowledgements

We thank the Cancer Prevention Research Institute of Texas (grant nos. RR160057 to R.F. and R1225 to G.D.) for their generous funding, as well as the National Institutes of Health (grant nos. F32GM116370 and K99GM123221 to M.K.D., R01GM067230 to G.D., R01AG055577 and R01NS056224 to I.B., R01DK118032 and R01DK099478 to D.M., R01DC015784 and R21NS104826 to J.M., R33CA235254 and R35GM133522 to R.F., and DP2MH119423 to R.T.). R.H. acknowledges support by the Collaborative Research Center SFB 1278 (PolyTarget, project C04) funded by the Deutsche Forschungsgemeinschaft. We are grateful to the Live cell imaging facility at UT Southwestern for access to the Zeiss Airyscan confocal microscope (supported by grant no. 1S10OD021684-01). We thank D. Saucier and J. Amatruda for providing the zebrafish specimens and E. Sapoznik for his assistance with computer-aided design.

Author contributions

T.C., K.M.D. and R.F. designed the research. T.C. and R.F designed and built the microscopes. T.C. and S.V. prepared the CAD files and operated the microscope. T.C., M.K.D., K.M.D., B-J.C, P.R. and R.F. performed image analysis, partially under guidance by G.D. R.H. simulated PSFs. W.M.W., C.N., H.Z., V.Z., M.M.M., E.J., C.H., D.M., I.B., H.Z., R.T., J.M. and S.M. provided specimens and guided imaging. T.C., K.M.D. and R.F. wrote the manuscript. All authors read and provided feedback on the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information is available for this paper at https://doi.org/10.1038/s41592-019-0615-4.

Correspondence and requests for materials should be addressed to K.M.D. or R.F.

Peer review information Rita Strack was the primary editor on this article and managed its editorial process and peer review in collaboration with the rest of the editorial team.

Reprints and permissions information is available at www.nature.com/reprints.

nature research

Corresponding author(s): Reto Fiolka
Kevin Dean

Last updated by author(s): Apr 10, 2019

Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see Authors & Referees and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☒	☐	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Labview control software that was developed by Coleman technologies.
Data analysis	ImageJ, Matlab

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

We have no restrictions on data availability. Data can be requested from the corresponding author.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Resolution measurements of our miASLM microscopes were conducted by averaging over 5 green fluorescent microspheres. Kidneys, olfactory nodes, expanded hippocampus slices, and bone marrow niches have been imaged multiple times (e.g., >5) for ongoing research efforts.
Data exclusions	No data were excluded intentionally.
Replication	Replicate experiments were successful
Randomization	not applicable
Blinding	not applicable

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study	
☐	☒	Antibodies
☒	☐	Eukaryotic cell lines
☒	☐	Palaeontology
☐	☒	Animals and other organisms
☒	☐	Human research participants
☒	☐	Clinical data

Methods

n/a	Involved in the study	
☒	☐	ChIP-seq
☒	☐	Flow cytometry
☒	☐	MRI-based neuroimaging

Antibodies

Antibodies used	chicken anti-GFP: GFP-1020, Aves Labs goat anti-c-kit: BAF1356, R&D Systems Alexa Fluor 488-AffiniPure F(ab')2 fragment donkey anti-chicken IgY, 703-546-155, JacksonImmuno Alexa Fluor 647-AffiniPure F(ab')2 fragment donkey anti-goat IgG, 703-546-147, JacksonImmuno
Validation	*Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.*

Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Laboratory animals	Danio Rerio (Zebrafish), Strain: wik with trangenic krdl:mCherry, sex: male and female, age: 3dpf. Olfactory bulb: Age - 3.32 month old male transgenic mouse derived from strain C57Bl/6J. Expanded hippocampus: 3 month old female wild-type B6 mouse. Kidney: 3 day old neonatal mouse, C57Bl/6 Bone Marrow: 6-8 week old female mouse (Jackson Laboratory 008320, LepR-Cre (Leprtm2(cre)Rck/J) Rs26-tdTomato (B6.Cg-Gt(ROSA)26Sortm9(CAG-tdTomato)Hze/J, JAX:007909) α-catulin-GFP (Ctnnal1tm1.1Sjm/J, JAX: 028342)
Wild animals	n/a
Field-collected samples	n/a
Ethics oversight	Mice and Zebrafish maintenance and experiments were conducted in strict accordance with UTSW and Columbia university IACUC guidelines. UTSW approved the study protocol and have complied with all relevant ethical regulations

Note that full information on the approval of the study protocol must also be provided in the manuscript.